Anshul **Nasery**

PhD Student, University of Washington

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Research Interests

I work on problems around the security, privacy and robustness of AI systems. My work is informed by thorough empirical analyses along with a systems-level view of practical security and privacy considerations. Some of my recent projects include analysing the causes and effects of memorization during pre-training of large language models, fingerprinting and watermarking LLMs, and adversarial attacks to preserve the privacy of personal images against AI-powered image editing.

Education

University of Washington Sept 2023 – Present
PhD in Computer Science. Advisor: Prof. Sewoong Oh.

Indian Institute of Technology, Bombay Jul 2017 – Aug 2021
Bachelor of Technology (With Honors) in Computer Science and Engineering, Minor in Statistics. GPA: 9.58/10.

Research Experience

Google Research, New York City Jun 2026 – Sept 2026
Research Intern | Host: Anton Tsitsulin

- Studied causes and effects of verbatim memorization in LLMs through controlled experiments, pre-training over 4,000 language models (70M–1.3B parameters), with varying data distributions, de-duplication pipelines and model shapes.
- Fit scaling laws to measure and predict how data pipelines and modelling choices affect memorization and generalization, and investigated strategies to mitigate performance drops.

University of Washington Sept 2023 – Present
Graduate Research Assistant

- **Robust and Scalable Fingerprinting for LLMs**: Investigated techniques for secure model sharing through model fingerprinting, leveraging memorization and backdoor attacks for large language models. Developed a novel sampling technique for backdoors balancing persistence and utility drop [NeurIPS'25 Spotlight]. Proposed a practical threat model and adaptive attacks achieving perfect attack success rates against 10 fingerprinting schemes [SaTML'26]. Further developed robust schemes based on watermark distillation for secure fingerprinting [ICML-W'25, Pre-print].
- **Protecting Images against Malicious AI-powered Edits**: Developed transferable adversarial attacks to trigger content moderation filters for commercial image editing systems, offering practical protection rates of over 90% against non-consensual image edits across OpenAI, Google and xAI's APIs, massively improving over previous approaches[Pre-print].
- **Model merging for efficient ensembles**: Extended Git Re-Basin to produce merged models of varying sizes. Proposed a novel feature adjustment step to distill knowledge into a merged model with minimal data requirements [CVPR'2025].

Microsoft Research Sept 2023 – Dec 2023
Research Collaboration | Hosts: Harkirat Singh, Vibhav Vineet

- Developed a training-free method for spatio-temporally controlling the outputs of any video generation model. Proposed an evaluation framework and benchmark for controllable video generation [CVPR'24].

Google Research, Bangalore, India Jul 2021 – Jul 2023
Pre-Doctoral Researcher | Advisors: Dr. Prateek Jain, Dr. Praneeth Netrapalli

- **Investigating and Mitigating Simplicity Bias**: Studied simplicity bias and out-of-distribution (OOD) generalization of neural networks, empirically analyzing the mechanisms for the former through controlled experiments on synthetic data. Proposed training algorithms to mitigate simplicity bias and boost OOD robustness of models [ICLR'23, ICML-W'22].
- **Inference Efficient Neural Networks**: Developed a novel regularizer to minimize FLOPs and on-device latency for vision and language models, achieving a 50% reduction in FLOPs on BERT with minimal performance drop [CVPR-W'24].

Publications

Security and Privacy

[S.1] **Scalable Fingerprinting of Large Language Models** [🔗]
Anshul Nasery, Jonathan Hayase, Creston Brooks, Peiyao Sheng, Himanshu Tyagi, Pramod Viswanath, Sewoong Oh
Spotlight *at 38th Conference on Advances in Neural Information Processing Systems* [**NeurIPS'25**]

[S.2] **Are Robust LLM Fingerprints Adversarially Robust?** [🔗]
Anshul Nasery, Edoardo Contente, Alkin Kaz, Pramod Viswanath, Sewoong Oh
IEEE Conference on Secure and Trustworthy Machine Learning [**SaTML'26**]

[S.3] **Towards Quantifying the Causes and Effects of Verbatim Memorization**
Anshul Nasery, Simon Geisler, Anton Tsitsulin
Under preparation

[S.4] **MIRAGE: Protecting against Malicious Image Editing via False Moderation** [🔗]
Anshul Nasery, Ramnath Kumar, Cho-Jui Hsieh, Sewoong Oh
Under Review [**arXiv, 2026**]

[S.5] **Language Model Fingerprinting Requires Rethinking Watermark Teachers** [🔗]
Jeongyeon Hwang, **Anshul Nasery**, Sewoong Oh, Jungseul Ok
Under Review [**arXiv, 2026**]

[S.6] **OML: Open, Monetizable, and Loyal AI** [🔗]
Zerui Cheng, Edoardo Contente, Ben Finch, Oleg Golev, Jonathan Hayase, Andrew Miller, Niusha Moshrefi, **Anshul Nasery**, Sandeep Nailwal, Sewoong Oh, Himanshu Tyagi, Pramod Viswanath
Spotlight Position Paper at 39th Conference on Advances in Neural Information Processing Systems [**NeurIPS'26**]

[S.7] **Label Differential Privacy via Aggregation** [🔗]
Anand Brahmbhatt, Rishi Saket, Shreyas Havaldar, **Anshul Nasery**, Aravindan Raghuveer
Pre-print [**arXiv, 2023**]

Robustness

[R.1] **Feature Reconstruction from Outputs Can Mitigate Simplicity Bias in Neural Networks** [🔗]
Sravanti Addepalli*, **Anshul Nasery***, R Venkatesh Babu, Praneeth Netrapalli, Prateek Jain
International Conference on Learning Representations, 2023 [**ICLR'23**]

[R.2] **DAFT: Distilling Adversarially Finetuned Teachers for better OOD generalization** [🔗]
Anshul Nasery, Sravanti Addepalli, Praneeth Netrapalli, Prateek Jain
Principles of Distribution Shifts Workshop, ICML 2022 [**ICML-W'22**]

[R.3] **Training for the Future: A Simple Gradient Interpolation loss to Generalize Along Time** [🔗]
Anshul Nasery*, Soumyadeep Thakur*, Vihari Piratla, Abir De, Sunita Sarawagi
34th Conference on Advances in Neural Information Processing Systems [**NeurIPS'21**]

[R.4] **Teaching CNNs to Mimic Human Visual Cognitive Process & Regularise Texture-Shape Bias** [🔗]
Satyam Mohla*, **Anshul Nasery***, Biplab Banerjee
IEEE International Conference on Acoustics, Speech and Signal Processing, 2022 [**ICASSP'22**]

Efficiency

[E.1] **PLeaS – Merging Models with Permutations and Least Squares** [🔗]
Anshul Nasery*, Jonathan Hayase*, Pang Wei Koh, Sewoong Oh
IEEE / CVF Computer Vision and Pattern Recognition Conference, 2025 [**CVPR'25**]

[E.2] **What if Neural Networks had SVDs?** [🔗]
Alexander Mathiasen, Frederik Hvilshøj, Jakob Rødsgaard Jørgensen, **Anshul Nasery**, Davide Mottin
Spotlight *at 33rd Conference on Advances in Neural Information Processing Systems* [**NeurIPS'20**]

[E.3] **End-to-End Neural Network Compression via $\frac{\ell_1}{\ell_2}$ Regularized Latency Surrogates** [🔗]
Anshul Nasery, Hardik Shah, Arun Suggala, Prateek Jain
Oral *at Mobile AI Workshop at CVPR 2024* [**CVPR-W'24**]

Applications

[A.1] **Peekaboo: Interactive Video Generation via Masked-Diffusion** [🔗]
Yash Jain*, **Anshul Nasery***, Vibhav Vineet, Harkirat Behl
IEEE / CVF Computer Vision and Pattern Recognition Conference, 2024 [**CVPR'24**]

[A.2] **MIMOQA: Multimodal Input Multimodal Output Question Answering** [🔗]
Hrituraj Singh, **Anshul Nasery***, Denil Mehta*, Aishwarya Agarwal, Jatin Lamba, Balaji Vasan Srinivasan
2021 Conference of the North American Chapter of the Association for Computational Linguistics [**NAACL'21**]

[A.3] **Rule Augmented Unsupervised Constituency Parsing** [🔗]
Atul Sahay*, **Anshul Nasery***, Ayush Maheshwari, Ganesh Ramakrishnan, Rishabh Iyer
Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021 [**Findings of ACL'21**]

* Equal contribution. Full publication record: Google Scholar.

Technical Skills and Frameworks

PyTorch, JAX, LLM pre-training, Evaluation

Invited Talks

Detecting and Preventing Model Misuse through Fingerprinting
- University of Tübingen (Bethge Lab) — March 2026
- Google Research India — Dec 2025
- CMU CyLab — Oct 2025

Service and Teaching

- **NeurIPS 2026 Top Reviewer.** Reviewer for ICML (2022, 2024), NeurIPS (2022–2026), ICLR (2022, 2023, 2025, 2026), CVPR (2023, 2024), COLM (2024).
- Teaching assistant for a graduate-level course on Advanced ML at the University of Washington, undergraduate courses on Artificial Intelligence, Machine Learning, and Quantum Mechanics at IIT Bombay.